THE RELATIVE LERAY SPECTRAL SEQUENCE FOR A COMPOSITION OF MORPHISMS OF RINGED TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's own argument: the spectral sequence is the standard Grothendieck one, and the author names the two companion results of the same chapter which supply the acyclicity hypotheses, so the application is complete.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Grothendieck spectral sequence for a composition of functors. Derived Categories, the Grothendieck spectral sequence attached to a composition of functors, together with the acyclicity hypotheses it requires. Supplies the spectral sequence asserted in the statement.
- Acyclicity of the pushforward of an injective sheaf and of totally acyclic sheaves. Same chapter, the companion results stating that the pushforward of an injective sheaf is totally acyclic and that totally acyclic sheaves are acyclic for the functors involved. Supplies the two acyclicity inputs which make the Grothendieck spectral sequence applicable.

Source reference. Original lemma 015N: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2336–2336. Navigation pointer invoked at proof line 2336; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 072Z: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2212–2212. Navigation pointer invoked at proof

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line 2317; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0731: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2273–2273. Navigation pointer invoked at proof line 2319; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1 (Relative Leray spectral sequence)**.** *Let $f : (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C}) \to (Sh(\mathcal{D}), \mathcal{O}_\mathcal{D})$ and $g : (Sh(\mathcal{D}), \mathcal{O}_\mathcal{D}) \to (Sh(\mathcal{E}), \mathcal{O}_\mathcal{E})$ be morphisms of ringed topoi. Let $\mathcal{F}$ be an $\mathcal{O}_\mathcal{C}$-module. There is a spectral sequence with*

$$E_2^{p,q} = R^p g_*(R^q f_* \mathcal{F})$$

converging to $R^{p+q}(g \circ f)_* \mathcal{F}$. This spectral sequence is functorial in $\mathcal{F}$, and there is a version for bounded below complexes of $\mathcal{O}_\mathcal{C}$-modules.

Proof. This is a Grothendieck spectral sequence for composition of functors, see Derived Categories, Lemma 015N and Lemmas 072Z and 0731. □